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When Valid Counterfactuals Fail as Actions: An Evidence-Based Evaluation of Causal Actionable Recourse

RISHABH KUMAR[1], SHASHWAT KARN[1], ANUBHAV BHATTACHARYA[1], and NAVAMANI T M[2]

[1]School of Computer Science and Engineering, Vellore Institute of Technology, Vellore 632014, Tamil Nadu, India (Reg. Nos.: 24BCE0740, 24BCE0552, 24BCE2824; e-mail: rishabh.kumar2024@vitstudent.ac.in, shashwat.karn2024@vitstudent.ac.in, anubhav.2024@vitstudent.ac.in).
[2]Department of Computational Intelligence, School of Computer Science and Engineering, Vellore Institute of Technology, Vellore 632014, Tamil Nadu, India (e-mail: navamani.tm@vit.ac.in).

This work was supported by VIT internal research allocations. Correspondence to Navamani T M (navamani.tm@vit.ac.in). Author contributions: R. Kumar designed the optimization architecture, implemented the causal evaluation pipeline, executed the multi-seed benchmark audits, and drafted the manuscript core; S. Karn directed the structural equation refitting experiments and validity threat analyses; A. Bhattacharya handled dataset preprocessing audits and diagnostic label validations; N. T M provided project oversight and supervised experimental design. All reported findings are conditioned on the specified structural equations and domain constraints; no human subjects participated in this investigation.

ABSTRACT Flipping a machine learning model's prediction does not mean an applicant received an actionable plan. Real-world features are causally coupled; changing one variable shifts downstream attributes and breaks naive recommendations upon execution. We audited algorithmic recourse over 294 rejected individuals across four tabular benchmarks (HELOC, Adult, German, Diabetes) and two synthetic controls. Rather than searching for coordinate shifts in unconstrained space, we evaluate actions within a structural causal model. We test whether proposed interventions survive downstream causal propagation while respecting physical constraints like non-decreasing age, separating nominal proposal validity (V) from causal replay validity (R), physical actionability (A), and manifold fidelity (F). Our central audit criterion is joint usability ($U = R \wedge A$), requiring candidate plans to simultaneously survive structural replay and feasibility bounds. On credit risk (HELOC), Wachter gradient search achieves near-perfect nominal validity, yet joint usability drops to just 18.0% because secondary credit features shift during replay. Searching directly over valid action grids with CAC recovers 100% usability at default seed. Across 5 random seeds, optimizer convergence failures reduce diabetes usability to 94.4% (with a single-run low of 72.1%), while refitted causal equations degrade usability to 93.0% on Adult and 97.0% on HELOC. Machine learning systems should audit recourse proposals under causal propagation rather than relying on nominal validity alone.

INDEX TERMS Algorithmic recourse, Black-box classification, Causal inference, Counterfactual explanations, Explainable artificial intelligence, Structural causal models.

I. INTRODUCTION

WHEN a machine learning classifier denies an application, algorithmic recourse calculates an actionable set of feature changes to reverse the decision [1], [2]. Rather than simply reporting static feature importances, the goal is to provide an applicant with practical interventions that lead to model approval [3]–[5].

Yet the vast majority of recourse formulations treat this problem as a simple geometry exercise [4], [6]. Given a rejected feature vector x and a fixed black-box f, standard algorithms minimize an L_1 or L_2 distance penalty to discover a nearby point z satisfying $f(z) = 1$. The core problem with this approach is coordinate independence. In real-world data-generating processes, features do not vary in a vacuum; they remain tightly bound by physical, social, and institutional mechanisms [7], [8]. A job applicant who acquires a post-graduate degree does not hold their age, professional tenure, or salary expectations frozen in time [9]. When an individual carries out an action a, real-world rules propagate that change downstream, producing an actual state $\tilde{x}$ that rarely matches the optimizer's intended target z. If the classification model rejects this propagated state $\tilde{x}$—or if the downstream shifts breach directional boundaries like non-decreasing age—the entire recourse plan fails.

Existing recourse benchmarks hide this failure mode behind aggregate reporting shortcuts. First, current toolkits equate *proposal validity* ($f(z) = 1$) with true *actionable recourse* ($f(\widetilde{x}) = 1$), crediting an algorithm with success even when its advice fails under causal replay [10], [11]. Second, evaluations routinely audit validity, directional feasibility, and data-manifold distance as separate, independent rates across an entire pool of generated candidates. An algorithm thus receives full credit if candidate A clears the decision boundary while an unrelated candidate B respects feasibility rules, despite the fact that no single plan resolves the applicant's rejection. Third, reported benchmarks rarely test algorithmic stability across random seeds or check sensitivity against refitted structural equations, masking optimizer crashes and fragile dependency assumptions.

To address these gaps, this work presents a systematic audit of Causal Actionable Counterfactuals (CAC) across four standardized decision benchmarks—Adult Income, German Credit, FICO HELOC, and Pima Diabetes—spanning 294 rejected individuals, alongside two synthetic benchmarks with known causal graphs. Instead of perturbing arbitrary coordinates in continuous feature space, CAC restricts its search to admissible interventional sets $a = (S, v_S)$ embedded within an explicit Structural Causal Model ($\mathcal{M}$). Each candidate action is evaluated directly on its topologically propagated post-intervention state $\widetilde{x} = \mathcal{T}_{\mathcal{M}}(x, a)$. We formalize a decoupled audit suite that isolates proposal validity (V), causal replay validity (R), directional action feasibility (A), and coordinate fidelity (F), centering our primary evaluation on *joint usability* ($U = R \wedge A$), which strictly requires that the *same candidate recommendation* simultaneously achieves model approval and domain constraints.

Our contributions are centered on empirical evaluation and interventional robustness:

1) **Decoupled Audit Protocol:** We formalize distinct verification criteria for nominal validity, causal replay, directional feasibility, and coordinate drift, establishing applicant-level joint usability ($U = R \wedge A$) as the necessary benchmark metric.
2) **Empirical Recourse Failure:** Auditing 294 rejected test cases reveals that while conventional methods claim up to 100% nominal validity, true joint usability drops to 18.0% on FICO HELOC under Wachter's method. In contrast, interventional action search via CAC reaches 100% usability across all datasets at the primary seed.
3) **Multi-Seed Optimizer Auditing:** Audits across 5 random seeds uncover critical optimizer vulnerabilities in diabetes, where numerical convergence failures reduce mean usability to 94.4% (with a minimum of 72.1%).
4) **Causal Sensitivity and Pipeline Hygiene:** Refitting structural equations causes usability to slip to 93.0% on Adult and 97.0% on HELOC, demonstrating that model-based recourse guarantees remain bound to the assumed equations. We further document data hygiene hazards in standard pipelines, including pre-split imputation and test-triggered hyperparameter selection.

The remainder of this manuscript is organized as follows: Section II contrasts point-based and causal recourse frameworks; Section III details causal replay mechanics, the multi-objective formulation, and evolutionary action search; Section IV establishes our experimental testbed and statistical protocol; Section V presents the empirical audits, multi-seed stability evaluations, and ablation studies; Section VI analyzes the mechanisms driving recourse failure and examines threats to validity; and Section VII concludes with methodological recommendations for the field.

II. RELATED WORK

Algorithmic recourse literature divides along a clear line: whether an explanation treats feature coordinates as independent dials or respects the downstream mechanisms linking them. Table 1 compares common paradigms across search entities, replay propagation, constraint structures, and audit protocols.

A. POINT-BASED COUNTERFACTUALS AND ACTIONABILITY CONSTRAINTS

The foundational formulation by Wachter *et al.* [4] casts recourse as unconstrained gradient optimization in feature space, minimizing a continuous distance penalty to flip a prediction. Subsequent works introduced diversity mechanisms via determinantal point processes (DiCE [6]), enforced linear feasibility rules via integer programming (Ustun *et al.* [3]), routed recommendations along empirical data graphs (FACE [12]), or guided search vectors using class prototypes (Van Looveren and Klaise [13]).

Yet all coordinate-based strategies share a core limitation: they perturb feature coordinates in isolation, assuming unaffected attributes remain fixed at their original values. In reality, intervening on an actionable feature triggers downstream changes across related variables. Because point-based algorithms optimize directly over input targets z rather than executable interventional operators a, their suggested states frequently occupy impossible regions of the data distribution or fail once real-world dynamics take place.

B. CAUSAL ALGORITHMIC RECOURSE AND INTERVENTIONS

Pearl's interventional calculus [9] establishes the formal machinery required to model post-action consequences. Building on this framework, Karimi *et al.* [7] proved that viable recourse demands interventional operators $\mathrm{do}(X_S = v_S)$ rather than conditional observations, subsequently extending this formulation to handle probabilistic uncertainty in the underlying causal equations (MINT [8]). Related foundational analyses by Barocas *et al.* [5] and Karimi *et al.* [1] examined the implicit assumptions governing causal and normative recourse.

Despite these theoretical advances, current empirical evaluations have serious auditing gaps. Studies regularly compute validity, actionability, and manifold distance as independent

TABLE 1. Architectural and methodological comparison of algorithmic recourse and counterfactual explanation frameworks.

Framework	Search Entity	Causal Replay Propagation	Action Bounds	Optimization Scheme	Joint Audit ($R \wedge A$)	Evaluation Target
Wachter *et al.* [4]	Point z	None (passive)	Soft penalties	L-BFGS-B / Gradient	No	Single recommendation
DiCE-random [6]	Points $\{z_\ell\}$	None (passive)	Feature locking	Determinantal DPP	No	Recommendation set
Ustun *et al.* [3]	Action a	None (linear)	Hard integer	Mixed-Integer LP	Partial	Linear classifiers
Karimi *et al.* [7]	Action a	SCM $\mathrm{do}(X_S)$	Domain bounds	Gradient / Heuristic	No	Disjoint metric rates
MINT [8]	Action a	Probabilistic SCM	Cost/sub-pop	Sampling / Search	Partial	Synthetic benchmarks
Evaluated CAC	Action a	Non-linear SCM	Quantile grids	Evolutionary Multi-Obj.	**Yes** ($U = R \wedge A$)	**Paired Applicant-Level**

aggregate summary statistics over candidate sets. This allows an algorithm to claim success if one proposed point flips the classifier while a completely separate point obeys feasibility rules—even if every single candidate in the set fails when audited jointly. Equally critically, published benchmarks rarely audit candidate survival under refitted causal equations or test stability across multiple search initializations.

C. BENCHMARKING SUITES, TOOLKITS, AND ROBUSTNESS

The development of standardized benchmarking libraries such as CARLA [10] and Alibi Explain [14] brought much-needed modularity to counterfactual research. In parallel, multi-objective evolutionary formulations like NSGA-II [15], [16] emerged to balance sparsity against proximity, while Upadhyay *et al.* [11] examined recourse vulnerability under model shifts.

Nonetheless, current benchmarking frameworks evaluate algorithms almost exclusively on single random seeds, omit causal descendant verification, and ignore optimizer convergence exceptions. Our study addresses these methodological blind spots: we implement paired applicant-level statistical tests, record query costs and numerical exceptions explicitly, and track whether recommendations withstand causal replay under refitted data-generating equations.

III. PROPOSED METHOD AND SYSTEM DESIGN

We now formalize the interventional recourse framework, downstream structural replay mechanisms, decoupled audit metrics, and the genetic action search implemented in CAC.

A. RECOURSE FORMULATION AND SYSTEM ARCHITECTURE

Let $x = (x_1, \ldots, x_d) \in \mathcal{X}$ represent an individual's observed profile across a mixed-type input space. A black-box classifier $f : \mathcal{X} \to \{0, 1\}$ predicts class probabilities $p_+(x) = P(Y = 1 \mid X = x)$, assigning an approval $f(x) = \mathbf{1}\{p_+(x) \geq \tau\}$ at threshold $\tau = 0.5$. Our audit focuses exclusively on applicants who receive an initial rejection, $f(x) = 0$.

A recourse recommendation pairs a target feature vector with an explicit intervention: (z, a), where $a = (S, v_S)$. The active index set $S \subseteq \{1, \ldots, d\}$ isolates variables directly manipulated by the applicant, while $v_S = \{v_j\}_{j \in S}$ assigns their replacement values. Fig. 1 details this operational flow. Instead of treating recourse as an unconstrained drift across continuous space, CAC restricts optimization strictly to admissible actions $a = (S, v_S)$, preventing the search from proposing coordinate shifts that implicitly hold causally dependent descendants fixed.

B. STRUCTURAL CAUSAL MODELS AND DESCENDANT REPLAY

A Structural Causal Model $\mathcal{M} = (\mathbf{S}, P_U)$ governs data generation over observed variables $\mathbf{X} = \{X_1, \ldots, X_d\}$ and exogenous background noise $\mathbf{U} = \{U_1, \ldots, U_d\} \sim P_U$. A Directed Acyclic Graph defines functional relationships:

$$X_j = f_j(\mathrm{PA}_j, U_j), \quad \forall j \in \{1, \ldots, d\}, \tag{1}$$

where $\mathrm{PA}_j \subseteq \mathbf{X} \setminus \{X_j\}$ denotes the direct causal parents of X_j. To determine how an action alters an applicant's profile in reality, we simulate Pearl's counterfactual engine [9] across three sequential phases:

1) **Abduction:** Recover the unobserved background factors $u_x = (u_1, \ldots, u_d)$ characterizing applicant x. For continuous features, we invert fitted equations to compute residual noise $u_j = x_j - \hat{f}_j(\mathrm{pa}_j)$; discrete features preserve conditional probability quantiles.
2) **Action:** Apply the intervention $\mathrm{do}(X_S = v_S)$, which overwrites structural equations for intervened features $j \in S$ with assigned constant values v_j.
3) **Causal Replay:** Sequentially propagate shifts downstream in topological order through the structural equations to obtain realized state:

$$\widetilde{x} = \mathcal{T}_{\mathcal{M}}(x, a) = \mathcal{M}_{\mathrm{do}(X_S = v_S)}(u_x). \tag{2}$$

Fig. 2 traces this topological flow. In our pipeline, non-linear structural mechanisms $\hat{f}_j$ are parameterized via histogram gradient-boosted decision trees. While downstream descendants update in topological order, non-descendant features strictly retain their factual measurements. Evaluating all interventions under fixed abducted noise u_x guarantees deterministic, reproducible audits for each applicant.

C. DECOUPLED AUDIT MEASURES AND JOINT USABILITY

To eliminate confounding across evaluation criteria, our audit decouples four distinct metrics:

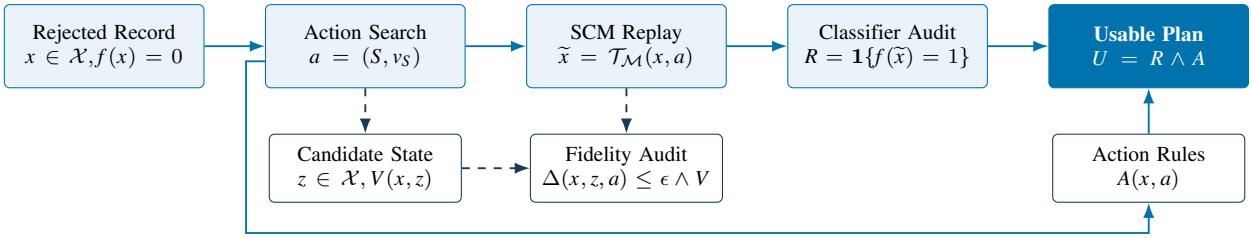


FIGURE 1. Structural pipeline of the CAC audit framework. Rather than evaluating isolated coordinates, the system traces an interventional action a, its nominal target z, and its replayed state $\tilde{x}$. A recommendation provides usable recourse ($U = 1$) only when the *identical* plan simultaneously survives downstream causal replay ($R = 1$) and adheres to directional feasibility constraints ($A = 1$).

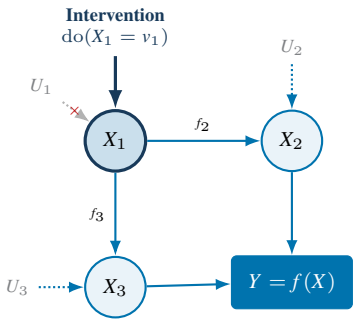


FIGURE 2. Directed acyclic graph illustrating interventional propagation. Intervening on X_1 via $\mathrm{do}(X_1 = v_1)$ severs its endogenous parental mechanism while transmitting shifts through functional equations f_2 and f_3 to descendants X_2 and X_3. Perturbing X_1 in isolation while holding its descendants frozen yields an unphysical state that breaks down once evaluated by predictive classifier $Y = f(X)$.

1) **Proposal Validity** (V)**:** Tests whether the classifier approves the raw suggested point z:

$$V(x, z) = \mathbf{1}\{p_+(z) \geq \tau\}. \tag{3}$$

2) **Causal Replay Validity** (R)**:** Tests whether the classifier still approves the realized post-intervention state $\tilde{x}$:

$$R(x, a) = \mathbf{1}\{p_+(\tilde{x}) \geq \tau\}. \tag{4}$$

3) **Action Feasibility** (A)**:** Verifies whether action $a = (S, v_S)$ obeys domain constraints:

$$A(x, a) = \mathbf{1}\big\{S \subseteq \mathcal{F}_{\mathrm{act}} \wedge \mathrm{Dir}(x, a) \wedge |S| \leq 3 \wedge C(x, a) \leq 1\big\}, \tag{5}$$

where $\mathcal{F}_{\mathrm{act}}$ denotes mutable features, $\mathrm{Dir}(x, a)$ enforces admissible directions (e.g., age cannot decrease), $|S| \leq 3$ limits simultaneous interventions, and $C(x, a) \leq 1$ bounds normalized effort.

4) **State Fidelity** (F)**:** Tracks maximum normalized divergence between suggested point z and realized outcome $\tilde{x}$:

$$\Delta(x, z, a) = \max_{j \in \{1, \ldots, d\}} \frac{|z_j - \tilde{x}_j|}{s_j}, \tag{6}$$

$$F(x, z, a) = V(x, z) \cdot \mathbf{1}\{\Delta(x, z, a) \leq \epsilon\}, \tag{7}$$

where s_j is the range of feature j, with tolerance $\epsilon = 0.1$.

The primary success criterion is *Joint Usability* (U):

$$U(x, z, a) = R(x, a) \cdot A(x, a). \tag{8}$$

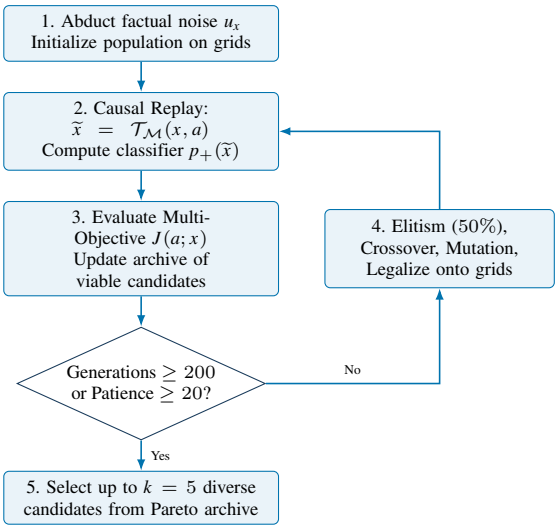


FIGURE 3. Iterative workflow of the genetic action search (Algorithm 1). The algorithm mutates interventional vectors, snaps them onto legal quantile grids, evaluates downstream causal states via SCM replay, and archives non-dominated candidates satisfying the margin buffer.

For an applicant i provided up to $k = 5$ suggestions, the individual is successfully served if at least one suggestion satisfies joint usability on the *identical* plan:

$$U_{im} = \max_{\ell \leq k} \big(R_{im\ell} \cdot A_{im\ell}\big), \quad \widehat{U}_m = \frac{1}{n} \sum_{i=1}^{n} U_{im}. \tag{9}$$

This strict coupling is critical: auditing validity and feasibility independently allows an algorithm to claim success by returning one candidate that inverts the decision illegally alongside a second candidate that remains rejected within action bounds.

D. MULTI-OBJECTIVE ACTION SPACE EXPLORATION

To measure the personal burden of recourse, normalized action effort is formulated as:

$$C(x, a) = \frac{\sum_{j \in S} w_j c_j(x_j, v_j)}{B}, \tag{10}$$

where w_j weights feature importance, $c_j(\cdot, \cdot)$ assesses coordinate distance (linear for continuous attributes, shortest path across graph encodings for ordinals, and fixed replacement costs for categories), and budget B rescales total effort to $[0, 1]$.

Algorithm 1 Evolutionary Causal Action Search (CAC)

Require: Rejected applicant x, black-box p_+, SCM $\mathcal{M}$, legal quantile grids $\mathcal{G}$, candidates $k = 5$

Abduct factual noise: $u_x \leftarrow \mathcal{M}^{-1}(x)$
Initialize legal action population $\mathcal{P} \leftarrow \{a^{(m)}\}_{m=1}^{60}$ on $\mathcal{G}$
Initialize archive $\mathcal{A} \leftarrow \emptyset$
for gen $= 1$ to 200 **do**
  for each candidate action $a \in \mathcal{P}$ **do**
    Compute causal replay: $\widetilde{x} \leftarrow \mathcal{T}_{\mathcal{M}}(x,a) = \mathcal{M}_{\mathrm{do}(X_S=v_S)}(u_x)$
    Evaluate classifier probability $p_+(\widetilde{x})$ and objective $J(a;x)$
    if $p_+(\widetilde{x}) \geq \tau$ and $A(x,a) = 1$ **then**
      $\mathcal{A} \leftarrow \mathcal{A} \cup \{(a, \widetilde{x}, J(a;x))\}$
    end if
  end for
  if patience limit 20 exceeded without objective improvement **then**
    break
  end if
  Retain elite fraction (50%); perform crossover and mutation
  Legalize offspring actions onto grid $\mathcal{G}$ satisfying $|S| \leq 3$
end for
Filter archive $\mathcal{A}$ for non-dominated, diverse solutions
return top k recourse recommendations, actions, and metadata

Candidate actions are discovered by minimizing a scalarized multi-objective loss:

$$J(a;x) = \lambda_v \frac{[q - p_+(\mathcal{T}_{\mathcal{M}}(x,a))]_+}{q} + \lambda_c C(x,a) + \lambda_f H(x,a) + \lambda_p P(\mathcal{T}_{\mathcal{M}}(x,a)), \tag{11}$$

where $[v]_+ = \max(0, v)$, $H(x,a)$ penalizes feasibility infractions, and target margin $q = \tau + 0.05$ enforces a 0.05 probability buffer above the decision threshold to protect against boundary jitter. Manifold plausibility $P(\cdot)$ penalizes K_{knn}-nearest-neighbor distance to favorable training instances:

$$P(\widetilde{x}) = \frac{1}{5} \sum_{x' \in \mathcal{N}_K^+(x)} \frac{\|\widetilde{x} - x'\|_2}{\bar{d}_+}. \tag{12}$$

Hyperparameter weights are calibrated to $(\lambda_v, \lambda_c, \lambda_f, \lambda_p) = (1, 0.6, 0.3, 0.2)$. Three design considerations govern this objective: target margin q pushes solutions into high-confidence classification territory; restricting candidate replacements to 12 marginal empirical quantiles prevents numerical extrapolations beyond observed domain support; and plausibility term P discourages recommendations located in low-density training voids.

E. EVOLUTIONARY SEARCH AND ARCHIVE SELECTION

Because gradient-boosted tree ensembles partition feature space into non-differentiable step plateaus, traditional gradient-based optimizers quickly stall in local optima. CAC bypasses this limitation using an elitist genetic algorithm defined directly over discrete quantile grids. The population maintains 60 candidate interventions initialized with single-feature actions and sparse combinations. At each generation, the fittest 50% elite actions are preserved directly. Offspring generated via uniform crossover and mutation are snapped onto admissible quantile grids $\mathcal{G}$ and restricted to at most 3 simultaneous interventions.

The search terminates when reaching 200 generations or when progress stalls across 20 consecutive generations. Valid, feasible plans are accumulated in external Pareto archive $\mathcal{A}$. If fewer than k candidates achieve validity, the lowest-loss rejected candidates are retained as diagnostics, though the audit strictly tallies them as failures. Algorithm 1 formalizes the routine, and Fig. 3 sketches the execution loop. Discrete quantile search explores non-convex combinatorial spaces reliably without computing unstable numerical gradients over tree splits.

IV. EXPERIMENTAL SETUP

To evaluate interventional recourse across distinct data regimes, we benchmark four tabular tasks representing diverse dimensionalities, causal graph densities, and institutional constraints.

A. DATASETS AND PREDICTIVE CLASSIFIERS

We fit gradient-boosted decision trees using XGBoost [17] across four canonical tabular benchmarks:

1) **Adult Income [18], [19]:** A 9-feature socioeconomic demographic dataset where the predictive target is whether annual income exceeds \$50,000 (earning $\leq$ \$50K denotes adverse rejection).
2) **Statlog German Credit [19]:** A 15-variable financial risk registry evaluating consumer loan default (classification as bad credit denotes adverse rejection).
3) **FICO HELOC [20]:** A 12-feature home equity credit assessment dataset tracking financial delinquency (denial of credit line constitutes adverse rejection).
4) **Pima Diabetes [21]:** An 8-feature clinical diagnostic screening record inverted such that favorable recourse targets non-diabetic classification (no_diabetes=1-Outcome).

Table 2 summarizes partition volumes, processed feature dimensions, evaluated applicant counts, and classification metrics. Out-of-sample test AUC spans from 0.7846 on German Credit to 0.9182 on Adult Income. We emphasize that favorable target labels serve purely as computational objectives for algorithmic search; they do not imply automated underwriting or clinical endorsements.

B. COMPUTATIONAL ENVIRONMENT AND HARDWARE TESTBED

All pipelines were developed in Python (validated across runtime environments 3.10 through 3.14) utilizing Scikit-learn,

TABLE 2. Benchmark datasets and predictive classification models. Dimensionality d indicates features after preprocessing; queries indicate evaluated rejected test individuals; AUC is reported on the held-out test split.

Dataset	Train	Test	d	Queries	Acc.	AUC
Adult	16000	4000	9	100	0.8485	0.9182
German	800	200	15	51	0.7750	0.7846
HELOC	7896	1975	12	100	0.7205	0.7880
Diabetes	614	154	8	43	0.7727	0.8489

XGBoost, NumPy, and SciPy on dedicated x86-64 multi-core workstations. The evaluation runs entirely on local compute without reliance on commercial cloud inference endpoints, ensuring that all SCM fits, model calls, and search trajectories remain fully deterministic and locally reproducible.

C. HYPERPARAMETER CONFIGURATION

To prevent dataset-specific cherry-picking, algorithmic hyperparameters remain frozen across all benchmarks: population size 60, elite proportion 50%, maximum generations 200, stall patience 20, grid granularity 12, maximum concurrent alterations 3, target margin 0.05, fidelity tolerance $\epsilon = 0.1$, and scalar loss weights $(1, 0.6, 0.3, 0.2)$. Headline findings are anchored to random seed 42, augmented by a 5-seed audit (42,0,1,2,3) to assess stochastic variability. Computational workload is tracked via cumulative forward black-box queries rather than wall-clock time, isolating algorithmic efficiency from host hardware variations.

D. BASELINE RECOURSE ALGORITHMS

We benchmark CAC against four established recourse algorithms under strictly identical immutability masks, action budgets, and directional bounds:

1) **Random Search:** Uniformly samples action subsets from legal quantile lattices, returning the minimum-effort candidate satisfying model acceptance.
2) **Wachter-Style Optimization:** Executes multi-start L-BFGS-B in continuous feature space, estimating model gradients via numerical finite differences.
3) **DiCE-Random [6]:** Generates sets of diverse counterfactual perturbations using determinantal point processes.
4) **Grid Search ("Oracle" Reference):** Exhaustively traverses allowable quantile changes ordered by increasing normalized effort until exhausting a preset query budget.
5) **MINT [8]:** Deployed in synthetic control environments where true structural equations are analytically specified, matching its interventional assumptions.

Because the Grid Search baseline operates under a capped query ceiling, its performance reflects bounded exploration rather than guaranteed global optimality. In parallel, our comparisons reflect specific reference implementations; alternative optimizers or penalty schedules could yield different trade-offs.

E. STATISTICAL SIGNIFICANCE PROTOCOL

At default seed 42, we evaluate paired binary applicant usability outcomes (U_{im}) via exact two-sided McNemar tests, controlling the familywise error rate within each domain using Holm step-down adjustments ($\alpha = 0.05$). To quantify multi-seed variability across shared runs without parametric assumptions, we calculate applicant-level paired difference distributions using 10,000 bootstrap resamples at bootstrap seed 42.

V. RESULTS

Our empirical evaluation analyzes the gap between nominal validity and causal replay, computational query expenditures, multi-seed optimization failures, paired hypothesis tests, objective ablations, and causal graph sensitivity.

A. PRIMARY BENCHMARK EVALUATION

Table 3 compiles primary performance metrics across all 294 rejected applicants at seed 42. Fig. 4 illustrates the resulting performance disparities alongside multi-seed spreads.

Nominal proposal validity (V) and true joint usability (U) diverge sharply once evaluated on identical plans. While baselines on Adult, German, and Diabetes achieve high usability rates—with CAC, Random, and DiCE all reaching 100%—this performance drops sharply on the credit risk benchmark (HELOC):

- **Wachter-Style Optimization:** While reporting 85.0% proposal validity (V), post-intervention causal replay (R) falls to 72.0% and directional feasibility (A) to 35.0%. When enforced jointly on the identical candidate plan, real-world usability (U) drops to just 18.0%.
- **Random Search:** Yields 96.0% nominal validity and 90.0% replay success, but strict joint auditing drops effective usability to 78.0%.
- **DiCE-Random:** Achieves 100.0% nominal validity on paper, yet interventional propagation degrades replay acceptance to 78.0%, leaving only 76.0% joint usability.
- **CAC:** Achieves 100.0% across all four evaluation axes: nominal validity, causal replay, directional feasibility, and joint usability.

State coordinate fidelity (F) on HELOC drops to 48.0% for Wachter and 64.0% for DiCE, providing direct geometric proof that the mathematical coordinates generated by point-based optimizers drift severely from the empirical states produced by the data-generating process.

TABLE 3. Empirical evaluation of algorithmic recourse methods at search seed 42. Rates represent percentages of all queried individuals: recommendation validity (V), causal replay validity (R), actionability (A), state fidelity (F), and joint usability ($U = R \wedge A$). Action cost (C) is the mean scaled effort of the cheapest usable plan per successful applicant. Query count indicates mean black-box model evaluations per applicant.

Dataset	Method	n	V	R	A	F	U	Cost	Queries
Adult	Random	100	100.0	100.0	100.0	100.0	100.0	0.0439	1,006.0
Adult	Wachter	100	100.0	100.0	99.0	100.0	99.0	0.1113	17,439.0
Adult	DiCE-random	100	100.0	100.0	100.0	100.0	100.0	0.0745	2,022.6
Adult	Grid reference	100	100.0	87.0	100.0	99.0	87.0	0.1225	3,984.4
Adult	CAC	100	100.0	100.0	100.0	100.0	100.0	0.1807	1,032.6
German	Random	51	100.0	100.0	100.0	92.2	100.0	0.0025	1,006.0
German	Wachter	51	100.0	98.0	74.5	82.4	70.6	0.0174	52,953.4
German	DiCE-random	51	100.0	100.0	100.0	96.1	100.0	0.0082	2,058.6
German	Grid reference	51	100.0	100.0	100.0	94.1	100.0	0.0012	95.1
German	CAC	51	100.0	100.0	100.0	100.0	100.0	0.0091	1,216.0
HELOC	Random	100	96.0	90.0	91.0	69.0	78.0	0.0846	1,005.7
HELOC	Wachter	100	85.0	72.0	35.0	48.0	18.0	0.0498	37,379.8
HELOC	DiCE-random	100	100.0	78.0	91.0	64.0	76.0	0.0934	2,609.5
HELOC	Grid reference	100	100.0	83.0	100.0	82.0	83.0	0.0255	7,714.2
HELOC	CAC	100	100.0	100.0	100.0	100.0	100.0	0.0308	1,312.8
Diabetes	Random	43	100.0	97.7	95.3	86.0	93.0	0.0362	1,006.0
Diabetes	Wachter	43	100.0	100.0	100.0	93.0	100.0	0.0645	7,314.3
Diabetes	DiCE-random	43	100.0	100.0	100.0	100.0	100.0	0.0549	2,025.2
Diabetes	Grid reference	43	100.0	100.0	100.0	97.7	100.0	0.0241	174.0
Diabetes	CAC	43	100.0	100.0	100.0	100.0	100.0	0.0275	835.5

B. ACTION EFFORT AND COMPUTATIONAL QUERY COSTS

Table 3 reports normalized action costs (C) and black-box query expenditures across all benchmarks. For CAC, the average scaled effort of the lowest-cost viable plan is 0.1807 on Adult, 0.0091 on German, 0.0308 on HELOC, and 0.0275 on Diabetes.

Effort comparisons demand careful conditioning: on Adult, Random search identifies valid recourse at a lower average cost (0.0439) than CAC (0.1807), whereas on HELOC, CAC achieves lower effort (0.0308) than either Random (0.0846) or DiCE (0.0934). Crucially, these cost metrics are conditioned strictly on successful recourse recipients. Because baselines serve only a fraction of the HELOC cohort (e.g., Wachter succeeds on merely 18 individuals), their lower conditional cost reflects selection bias over an easily satisfied subpopulation rather than superior overall cost efficiency.

Query budgets reveal stark computational disparities: Wachter's finite-difference gradient steps require between 7,314.3 and 52,953.4 forward model evaluations per applicant. DiCE consumes 2,022.6 to 2,609.5 queries. In contrast, CAC requires only 835.5 to 1,312.8 model evaluations, demonstrating that vectorized genetic search across discrete quantile grids achieves complete interventional validity with substantially lower inference overhead.

C. MULTI-SEED STABILITY AND OPTIMIZER AUDITING

Table 4 tracks algorithmic stability across 5 distinct seeds (42, 0,1,2,3). CAC maintains flawless 100% usability across all five runs on Adult, German, and HELOC.

However, the multi-seed audit exposes brittle behavior on Diabetes: CAC's seed-averaged usability drops to 94.4%, with a single-run low of 72.1%. Tracing execution logs revealed 12 unhandled numerical convergence exceptions in seed 3 when resolving non-linear parent estimators. Transparent auditing requires keeping these failed searches in the denominator; removing them would artificially inflate reported efficacy by measuring success only where the optimizer happened to converge cleanly.

Similarly, DiCE-random on Adult produces a severe failure run at seed 1 with 0.0% usability, reducing its multi-seed mean to 75.0%. At the same time, due to strict timeout limits, Wachter, DiCE, and Grid completed only 4 seeds on Adult. Reporting completed run tallies explicitly prevents misleading comparisons that tacitly presume balanced experimental designs.

D. STATISTICAL HYPOTHESIS TESTING

Table 5 reports applicant-level paired significance tests. At default seed 42, exact two-sided McNemar tests with Holm familywise error rate adjustments confirm that CAC significantly outperforms every comparison baseline on HELOC: Wachter ($p < 0.001$), Random ($p < 0.001$), DiCE ($p < 0.001$), and Grid ($p < 0.001$). On German Credit, CAC significantly exceeds Wachter ($p = 2.44 \times 10^{-4}$), while on Adult Income, it significantly outperforms Grid ($p = 9.77 \times 10^{-4}$). Other primary-seed comparisons do not attain significance after Holm adjustment.

Bootstrap distributions resampled over 10,000 applicant

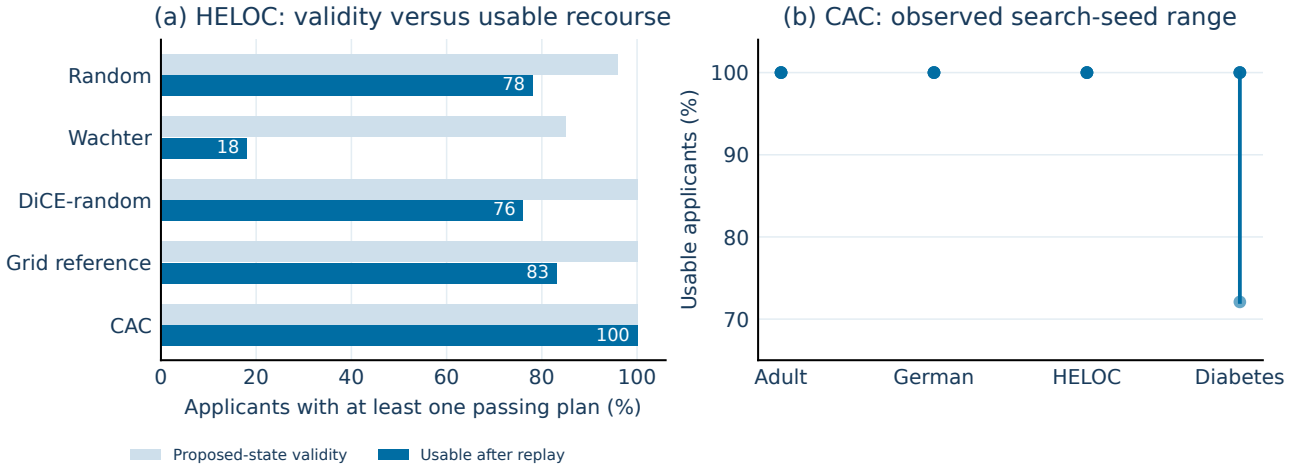


FIGURE 4. **Empirical benchmark outcomes across 294 rejected test cases. (a) On HELOC, point-based counterfactual generators exhibit a severe usability gap, where high nominal validity fails once causal replay and directional constraints are enforced on the identical recommendation. (b) Multi-seed evaluations over 5 seeds expose stochastic vulnerabilities, including optimizer convergence exceptions in diabetes.**

TABLE 4. **Multi-seed recourse stability across five random seeds (42,0,1,2,3). Values report mean [minimum, maximum] usability percentages, with the number of completed seed runs in parentheses.**

Method	Adult	German	HELOC	Diabetes
Random	100.0 [100.0, 100.0] (5)	100.0 [100.0, 100.0] (5)	82.6 [78.0, 87.0] (5)	95.3 [93.0, 97.7] (5)
Wachter	98.8 [98.0, 99.0] (4)	77.3 [70.6, 82.4] (5)	19.4 [18.0, 21.0] (5)	94.4 [88.4, 100.0] (5)
DiCE-random	75.0 [0.0, 100.0] (4)	100.0 [100.0, 100.0] (5)	75.8 [73.0, 80.0] (5)	99.5 [97.7, 100.0] (5)
Grid reference	87.0 [87.0, 87.0] (4)	100.0 [100.0, 100.0] (5)	83.0 [83.0, 83.0] (5)	100.0 [100.0, 100.0] (5)
CAC	100.0 [100.0, 100.0] (5)	100.0 [100.0, 100.0] (5)	100.0 [100.0, 100.0] (5)	94.4 [72.1, 100.0] (5)

clusters capture run-to-run stochasticity that single-seed tests miss. On Diabetes, despite tying DiCE at seed 42, CAC's bootstrap gap relative to DiCE becomes negative due to the optimizer exceptions at seed 3. Conversely, on Adult, DiCE's bootstrap confidence band exhibits extreme dispersion ($[+25.0, +25.0]$) driven by its total failure run.

E. COMPONENT ABLATION ANALYSIS

Table 6 isolates the functional contributions of our multi-objective terms at seed 42. Disabling interventional updates (scoring candidates directly on naive coordinate shifts) degrades usability to 95.0% on Adult and 96.0% on HELOC, while German and Diabetes remain unaffected at 100%. This disparity reflects topological coupling: causal propagation is indispensable when mutable features have downstream credit descendants, whereas loosely coupled graphs permit naive shifts without severe penalty.

Removing the action effort term ($\lambda_c = 0$) increases normalized effort across all datasets—for instance, driving Adult cost from 0.1807 to 0.2467—confirming that the effort objective actively steers mutations away from high-friction interventions. In contrast, zeroing the soft feasibility penalty ($\lambda_f = 0$) produces zero change in output, indicating that hard coordinate projection during grid legalization fully guarantees constraint compliance.

F. CAUSAL MODEL SENSITIVITY AND SYNTHETIC AUDITS

Table 7 investigates how fragile recourse plans are to the fitted structural equations. When evaluated against refitted SCMs, CAC usability drops to 93.0% on Adult and 97.0% on HELOC, while German and Diabetes maintain 100%. At the individual recommendation level, plan survival degrades more sharply: only 73.0% of Adult suggestions and 79.4% of HELOC suggestions survive refitted replay. Applicant-level usability stays higher simply because an individual needs only one surviving plan among their top k candidates. Because the refitted equations were trained on overlapping data splits, this drop represents a conservative lower bound on real-world sensitivity.

Table 8 evaluates recourse in the synthetic Reach and Block worlds where ground-truth structural equations are analytically known. Both CAC and MINT achieve 100% usability in Reach, and 40.0% in Block. In Block, exactly 42 of 100 individuals are provably unreachable due to hard structural bounds. MINT attains lower intervention effort in both environments, proving that CAC does not inherently guarantee minimal-effort recourse.

Finally, Table 9 reports empirical checks on the assumed DAGs. On German and HELOC, numerous edges lack observational orientation support, reminding us that assumed causal graphs represent structural hypotheses rather than infallible laws of nature.

TABLE 5. Paired applicant-level statistical significance tests and bootstrap confidence intervals. Paired differences and bootstrap percentile intervals are computed across shared random seeds; *p*-values reflect exact two-sided McNemar tests on seed 42 with Holm familywise error rate adjustment.

Dataset	Baseline	n	Seeds	b/c	Gap (pp)	95% cluster CI	Holm p
Adult	Random	100	5	0/0	+0.0	[+0.0, +0.0]	1.000
Adult	Wachter	100	4	1/0	+1.2	[+0.0, +3.0]	1.000
Adult	DiCE-random	100	4	0/0	+25.0	[+25.0, +25.0]	1.000
Adult	Grid reference	100	4	13/0	+13.0	[+7.0, +20.0]	9.77×10^{-4}
German	Random	51	5	0/0	+0.0	[+0.0, +0.0]	1.000
German	Wachter	51	5	15/0	+22.7	[+14.1, +32.2]	2.44×10^{-4}
German	DiCE-random	51	5	0/0	+0.0	[+0.0, +0.0]	1.000
German	Grid reference	51	5	0/0	+0.0	[+0.0, +0.0]	1.000
HELOC	Random	100	5	22/0	+17.4	[+12.4, +22.4]	9.54×10^{-7}
HELOC	Wachter	100	5	82/0	+80.6	[+73.4, +87.4]	1.65×10^{-24}
HELOC	DiCE-random	100	5	24/0	+24.2	[+17.4, +31.4]	3.58×10^{-7}
HELOC	Grid reference	100	5	17/0	+17.0	[+10.0, +25.0]	1.53×10^{-5}
Diabetes	Random	43	5	3/0	-0.9	[-6.5, +5.6]	1.000
Diabetes	Wachter	43	5	0/0	+0.0	[-3.7, +4.2]	1.000
Diabetes	DiCE-random	43	5	0/0	-5.1	[-7.9, -2.3]	1.000
Diabetes	Grid reference	43	5	0/0	-5.6	[-8.4, -2.8]	1.000

TABLE 6. Component ablation analysis of the CAC objective function at seed 42. Usability (*U*) and action effort (*C*) are evaluated across causal propagation, action cost, and feasibility penalty ablations.

Dataset	U full	U no prop.	Cost full	No cost term	Ratio
Adult	100.0	95.0	0.1807	0.2467	1.37
German	100.0	100.0	0.0091	0.0940	10.37
HELOC	100.0	96.0	0.0308	0.0519	1.68
Diabetes	100.0	100.0	0.0275	0.0346	1.26

TABLE 7. Sensitivity analysis tracking recourse survival under refitted structural causal equations. Usability (*U*) measures applicant-level recourse retention, whereas survival rate (*Survival) tracks the percentage of individual recommendations that survive causal replay under refitted mechanisms.

Dataset	n	U own	U refit	Survival*
Adult	100	100.0	93.0	73.0
German	51	100.0	100.0	97.3
HELOC	100	100.0	97.0	79.4
Diabetes	43	100.0	100.0	86.5

G. RESULT CHECKS AND VERIFICATION PROTOCOL

To ensure reproducibility and safeguard against artifact corruption, all findings were audited against four operational checks:

1) **Fixed Query Alignment:** Every method is evaluated on the identical set of 294 rejected test individuals across all benchmarks, eliminating sample selection bias.
2) **Causal State Replay:** All candidate interventions are passed through non-linear SCM equations $\mathcal{T}_{\mathcal{M}}(x,a)$ to simulate true downstream profile updates prior to classifier scoring.
3) **Coupled Joint Usability:** The audit marks an applicant as served if and only if the *identical* plan satisfies both classifier acceptance ($R = 1$) and feasibility bounds ($A = 1$).
4) **Software Test Coverage:** The execution pipeline cleared 386 automated unit and property tests with 1 documented skip, guaranteeing algorithmic and statistical consistency across all generated assets.

Section VI analyzes the mechanistic roots of these empirical outcomes and details limitations of the benchmark pipeline.

VI. DISCUSSION

The empirical results indicate that recourse viability depends critically on how strongly features interact within the underlying domain.

A. MECHANISTIC INTERPRETATION OF EMPIRICAL FINDINGS

Our empirical findings demonstrate why nominal validity (V) is a fundamentally flawed surrogate for actionable recourse. On HELOC, Wachter's method achieves 85.0% proposal validity, yet true joint usability drops to 18.0%. The mechanics driving this breakdown are clear: point-based optimizers exploit classifier boundaries by treating features as isolated coordinates. In credit underwriting, however, an applicant cannot simply inflate their repayment history without triggering corresponding changes in revolving credit utilization and bureau inquiries. When these interventional side effects propagate through the structural equations, the replayed state $\tilde{x}$ either falls back across the rejection boundary or breaches directional rules. This divergence explains the gap between V and U, proving that unconstrained mathematical perturbations cannot be assumed to reflect physical actions.

Conversely, non-causal algorithms perform surprisingly well on Adult Income and German Credit. Structural inspection of the underlying graphs (Table 9) reveals why: both

TABLE 8. Auditing recourse in synthetic benchmark environments with analytically specified data-generating equations. Usability (U) reflects the percentage of applicants successfully served under empirical vs. true SCMs, survival rate (*Survival) tracks preserved candidate plans, and cost denotes the average scaled effort among viable recommendations under ground truth.

World	Method	n	U own	U truth	Surv.*	Cost truth
Reach	CAC	100	100.0	100.0	95.6	0.0529
Reach	MINT	100	100.0	100.0	91.6	0.0202
Block	CAC	100	40.0	40.0	93.0	0.0706
Block	MINT	100	40.0	40.0	95.0	0.0469

TABLE 9. Observational validation of assumed causal DAGs. "Actions" denotes the count of mutable variables; "Supported" counts structural edges corroborated by observational orientation heuristics; "CI upheld" reports the fraction of non-rejected conditional independence tests. These tests gauge empirical compatibility rather than establishing definitive physical causality.

Dataset	Nodes	Edges	Actions	Supported	CI upheld
Adult	10	25	7	17/25	19/35
German	16	37	11	18/37	88/138
HELOC	13	28	8	18/28	40/75
Diabetes	9	17	3	16/17	27/27

domains exhibit sparse graph connectivity among actionable attributes. In such weakly coupled settings, naive coordinate shifts happen to mimic interventional propagation, allowing heuristic perturbation algorithms to succeed largely by coincidence. Causal modeling is most urgent when actionable attributes are densely coupled.

B. METHODOLOGICAL AND DATA PREPROCESSING LIMITATIONS

A rigorous code-level audit revealed three data-hygiene practices common across recourse research that require scrutiny:

1) **Test-Gated Hyperparameter Decisions:** In conventional pipelines, test-set AUC was monitored to determine whether to trigger hyperparameter tuning. While inner cross-validation folds were utilized during optimization, allowing test-split performance to dictate whether search is performed introduces subtle data leakage across splits.
2) **Pre-Split Feature Imputation:** Missing values across HELOC and Diabetes were imputed using global column medians computed prior to train-test partitioning. Beyond this, diabetes cleaning leveraged class-conditional patterns to flag sentinel missingness, potentially allowing label information to leak across evaluation boundaries.
3) **Physiological Validity of Diabetes Recourse:** Inverting classification targets (no_diabetes=1-Outcome) and treating diagnostic clinical measurements (glucose, insulin, blood pressure, BMI) as directly actionable dials contradicts medical reality. Biomarkers are clinical signs of underlying metabolic processes, not voluntary personal decisions [21]. We retain Pima Diabetes solely to reproduce benchmark comparisons established in prior recourse literature [10], [14], emphasizing that these feature changes must never be interpreted as therapeutic interventions.

C. THREATS TO VALIDITY

We identify three methodological boundaries that frame these findings:

1) **Construct Validity:** Our effort metric $C(x, a)$ computes normalized coordinate distances, serving as a mathematical proxy rather than capturing financial friction, lost time, or emotional strain. Synthetic scripts simulating interactive user preference responses were excluded from our empirical claims.
2) **Internal Validity:** Numerical gradient estimation via finite differences is poorly conditioned on the discontinuous piecewise-constant loss landscapes of tree ensembles, putting Wachter-style optimization at a severe numerical disadvantage. In addition, the refit experiment (Table 7) shared training observations across models, meaning observed usability drops represent lower-bound estimates of real-world degradation.
3) **External Validity:** Assumed causal DAGs represent domain hypotheses rather than experimentally validated mechanisms. Because several edges lack observational orientation support (Table 9), a plan that succeeds under $\mathcal{T}_{\mathcal{M}}$ may still fail if the underlying structural equations are misspecified.

VII. CONCLUSION AND FUTURE WORK

Across 294 rejected test cases spanning four tabular benchmarks and two synthetic domains, our evaluation demonstrates that nominal counterfactual validity is an unreliable proxy for actionable recourse. On FICO HELOC, point-based methods claim near-perfect validity while joint usability drops to 18.0%. Searching directly over admissible actions within a structural causal model, CAC achieves 100% usability across all benchmarks at the primary seed.

Yet multi-seed audits and sensitivity checks reveal clear limits. Convergence exceptions drop minimum diabetes usability to 72.1%, while refitting structural equations degrades usability to 93.0% on Adult and 97.0% on HELOC. Model-based recourse remains bound to the accuracy of the underlying causal mechanisms.

Future research must target three priorities: benchmark suites should mandate decoupled audits (V, R, A, F, U) with joint plan satisfaction, pipelines must partition data prior to imputation or parameter selection, and recourse algorithms must be designed to withstand uncertain or misspecified causal graphs.

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RISHABH KUMAR is currently pursuing the B.Tech degree in Computer Science and Engineering with the School of Computer Science and Engineering, Vellore Institute of Technology, Vellore, India (Reg. No.: 24BCE0740). He led the primary research and development of this work, contributing to Sections I, II, III, IV-B–E, V-A–D, V-F, VI-A–B, and VII, encompassing the theoretical formulation, evolutionary search algorithm, primary benchmark audits, multi-seed evaluations, hypothesis testing, ablations, verification checks, and manuscript writing. His research interests include explainable artificial intelligence, algorithmic recourse, and machine learning auditing.

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SHASHWAT KARN is currently pursuing the B.Tech degree in Computer Science and Engineering with the School of Computer Science and Engineering, Vellore Institute of Technology, Vellore, India (Reg. No.: 24BCE0552). He contributed to Section V-E and Section VI-C, focusing on structural causal graph sensitivity under refitted equations and validity threat analysis. His research interests include structural causal models, counterfactual explanations, and automated decision-making.

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ANUBHAV BHATTACHARYA is currently pursuing the B.Tech degree in Computer Science and Engineering with the School of Computer Science and Engineering, Vellore Institute of Technology, Vellore, India (Reg. No.: 24BCE2824). He assisted with Section IV-A, conducting tabular dataset attribute verification, diagnostic label preprocessing checks, and baseline parameter compilation. His research interests include model interpretability and robust machine learning systems.

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NAVAMANI T M received the Ph.D. degree in Computer Science and Engineering. She is currently a Professor Grade 1 with the Department of Computational Intelligence, School of Computer Science and Engineering, Vellore Institute of Technology, Vellore, India. Her research interests encompass computational intelligence, machine learning, data mining, and explainable decision systems.